

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5116

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 20%

QUESTIONS: 5

Total Marks:50

Annexure A

CONCEPT MAPPING

Code of Conduct:

(192325096)

I V. Sandeep Reddy (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

V. Sandeep Reddy
Signature of the student:

CONCEPT MAPPING

1. Create a concept map linking the following: Security Attacks, Threats, Risks and Security Goals (Confidentiality, Integrity, Availability). Perform the following tasks:
 - a. Show how each attack leads to specific threats and risks
 - b. Map how security goals help mitigate them Provide one real-world example for each link
2. Develop a concept map comparing substitution techniques: Caesar Cipher, Monoalphabetic Cipher, Vigenère Cipher and Hill Cipher. Perform the following tasks to connect based on: Key usage, Complexity and Security strength. Analyze which technique is most secure and why.
3. Construct a concept map for Classical Encryption Schemes: Substitution Techniques and Transposition Techniques. Perform the following tasks:
 - a. Show differences and similarities
 - i. Map how each affects the Data structure and Security level
 - ii. Analyze which is more vulnerable to attacks and why
4. Create a concept map connecting Steganography concepts: Data Hiding, Watermarking and Survivability. Perform the following tasks:
 - a. Show relationships between these concepts
 - b. Analyze how each contributes to secure communication
 - c. Include one real-world application for each
5. Draw a concept map linking number theory concepts used in cryptography: Modular Inverse, Extended Euclid Algorithm, Fermat's Little Theorem, Euler's Phi Function and Euler's Theorem. Perform the following tasks:
 - a. Show how each concept is mathematically connected
 - b. Map their role in encryption/decryption
 - c. Analyze which concept is most critical for public key cryptography and justify

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding ; limited or partial integration	Poor understanding ; unable to integrate concepts
Experimental Design	30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

a) Classical Encryption Schemes

Textual Concept Map Structure

Classical Encryption Schemes

Substitution Techniques → Replaces elements

Transposition Techniques → Rearranges elements

Both form the foundation of modern cipher design

a) Differences and Similarities

Features	Substitution	Transposition
Core Mechanism	Changes the identity of the characters	Changes the position of the characters.
Plaintext letters	Replaced entirely. The original letters are absent in the ciphertext.	Retained entirely. The ciphertext is an anagram of the plaintext.

i) Impact on Data Structure and Security Level

Substitution Techniques:-

- Data structure: The spatial order and length of the message remain identical but the internal 'alphabet' is complicated.
- Security level: Low to moderate. Monoalphabetic substitution is extremely weak. Polyalphabetic substitution raises the security level by flattening frequency distributions but is still breakable with sufficient text.

• Transposition Techniques:-

- Data structure:- The internal alphabet remains exactly the same as the original language, but the spatial arrangement
- Security level:- Low, Because it perfectly preserves language statistics. it simply requires geometric or mathematical un-scrambling to solve.

ii) Vulnerability Analysis: which is more vulnerable?

monoalphabetic substitution: is technically the most vulnerable to immediate attack due to its susceptibility analysis.

• why: in any language certain letters appear more frequently because monoalphabetic substitution maps one plaintext letter to exactly one ciphertext letter, the ciphertext perfectly inherited the statistical footprint of the plaintext language.

An attacker simply counts the most common symbols.

concept Map: Steganography concepts

Textual concept Map Structure

Steganography

- Data Hiding: focuses on concealing the existence of a secret payload.
- Water marking: focuses on asserting ownership or identity of the host medium
- Survivability: The required robustness that ensures both data hiding and watermarks withstand distortion.

a. Relationships Between The Concepts

- Survivability as The Anchor:- Both techniques completely on survivability, if a hidden message is destroyed when an image is compressed the data hiding has failed

Concept	Security Contribution	Realworld Application
Data Hiding	Covertiness: conceals the very existence of a communication channel	Activists embedding text documents in social media photos to bypass ship.
Watermarking	Authentication: proves ownership and prevent unauthorized tampering	Invisible Digital Rights management signatures in stock photography
Survivability	Resilience: Guarantees embedded data withstands transmission	Patient data remaining securely attached to medical X-rays even when resized.